A Dive Into Material Properties and Their Significance

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ME-2220 Section Four

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1. Overview

The world is constructed with materials, all designed to fill their designed role. This report will investigate the processes and procedures for measuring and calculating the limits of materials, as well as discuss different types of materials along with their respective strengths and weaknesses. The strengths and weaknesses of materials are called material properties, and are the deciding factors when determining the best materials to build with. This report will document two different types of materials, aluminum, and carbon fiber. Aluminum is a metal commonly used when a lightweight material is needed, that still possesses strength. Carbon fiber is a composite, lightweight, strong material. A composite is made from two or more different materials, creating a new material with unique properties. Carbon fiber is composed of layers of carbon that are mixed with binding agents to create a uniform stacked structure. The orientation of the layers will greatly affect the properties of the final material. Due to the difference in properties of carbon fiber based on the creation process, two samples will be tested. Both materials will be tested using a tension testing machine, which will pull the samples apart and measure how far they stretch. Using this, stress and strain can be calculated. Stress is defined as the force per unit area, and strain is the measured deformation of a material in response to stress. These values are crucial to further analysis of the properties of these materials. The process and procedures of the experiment, as well as the results will be discussed in the following pages.

2. Experimental Methods

The experiment was divided into two sections. Day one included the testing of the aluminum sample, and day two tested the carbon fiber sample. Both samples were tested to the American Society of Testing and Materials (ASTM) standards, specifically ASTM E8/E8M - 11. Using these standards is important for comparison's sake. If all tests keep the same constants such as sample cross sectional area, then comparisons can be directly drawn due to the only difference in samples being the material they are made from.

Testing was done on four aluminum samples, all with the same type of material. This gave four sets of similar values, which can be used to approximate how most aluminum will behave under these

conditions. All four aluminum samples failed in the same manner. Once the sample left the elastic region, necking could be seen, where the sample begins to stretch out, leaving a noticeably smaller cross sectional area. Next, the sample broke, with a ductile fracture. This is different from a brittle fracture, which has no visible necking.

In the next experiment, two samples of carbon fiber were tested. One with an arrangement of layers to be parallel with the direction of force, which will be referred to as 0|0|0|0. The other samples fibers are perpendicular to the direction of force, and will be referred to as 90|90|90|90. Due to the higher complexity in creation, as well as the multiple materials that compose it, carbon fiber has multiple modes of failure. The sample may experience fibers breaking, microcracks, debonding between the fibers, as well as delamination or separation between the different layers of laminate. Carbon fiber is a brittle material, and will not possess the same type of failure mode as aluminum. There will be no visible necking, and it will snap without warning when one of the failure modes described above occurs. The results for both samples will be shown in the next section.

3. Experimental Results

For the aluminum sample, strain was plotted, and stress was calculated using the cross sectional area, as well as the force applied. This can be seen with the purple line below. The line has many distinct regions, with the first being the elastic region. This linear line signifies how much stress and strain the material can withstand, while still being able to return to its original shape. The point where the graph turns is the yield point. This is where the elastic, and inelastic regions meet. This is the point of no return, where any amount of stress will cause permanent deformation in the sample. The plastic region continues to climb in stress, but at a higher cost of strain when compared to the elastic region. The peak of the plastic region is the ultimate strength, and is the maximum amount of stress a material can withstand without breaking. After the point of ultimate strength, the sample begins necking, and will continue until fracture. Finally, the toughness can be calculated as the area underneath the curve, and is done using trapezoidal riemann sums, which break the sample into small trapezoids and sum all of the areas.

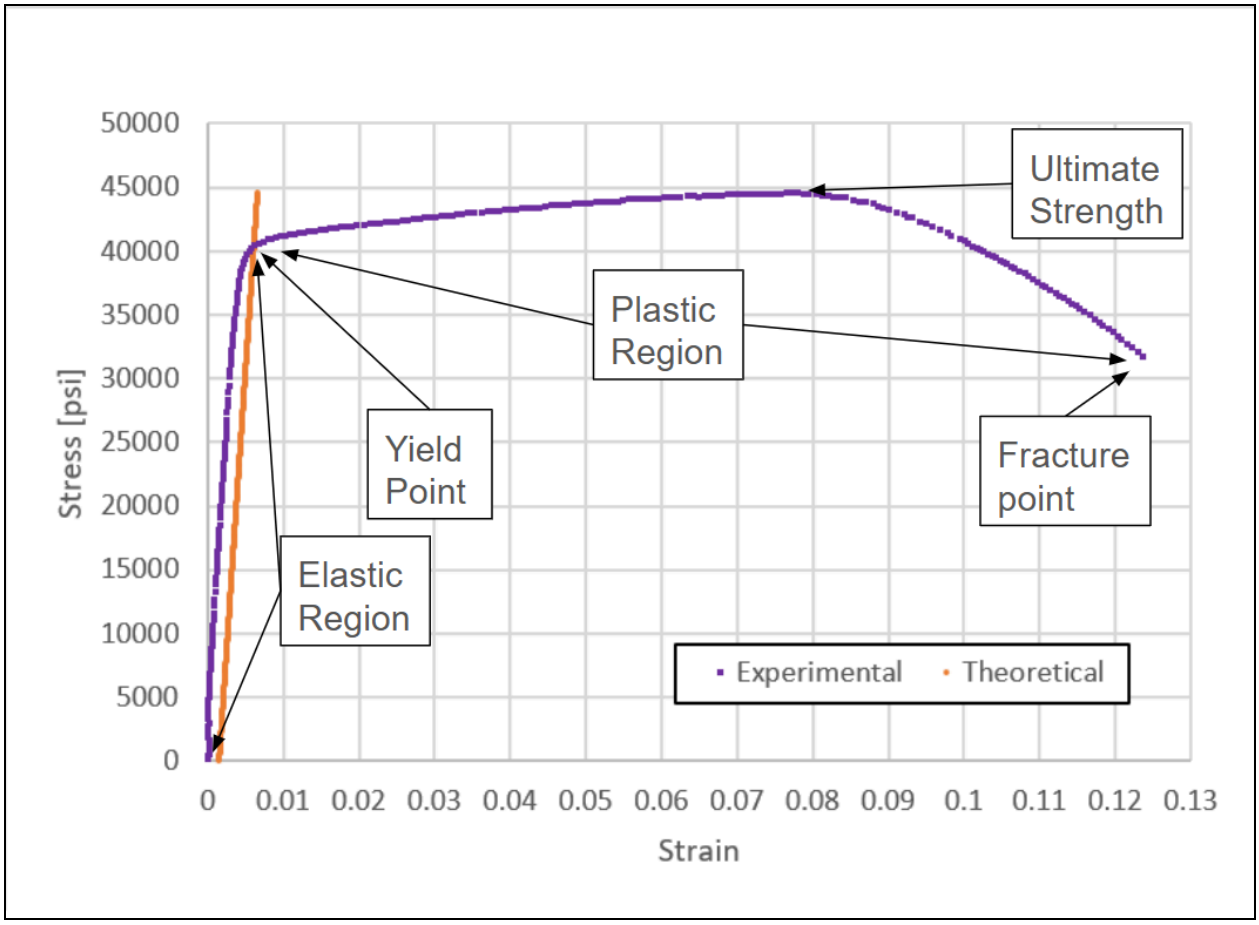


Figure 1: Aluminum Stress vs Strain with 0.2% offset

The stress strain plot for the carbon fiber samples looks different to the aluminum plot. This is mostly due to the difference in failure methods between the two. Aluminum's ductile failure is an elongated method of failure which is visible on the graph, where the brittle nature of carbon fiber leads to an abrupt failure of both carbon fiber samples. The main difference between the aluminum and carbon fiber plots, is the lack of a plastic region in the carbon fiber graph. This means that the Ultimate strength, yield strength, and fracture points, all share the same point. Toughness is calculated using the same methods as the aluminum.

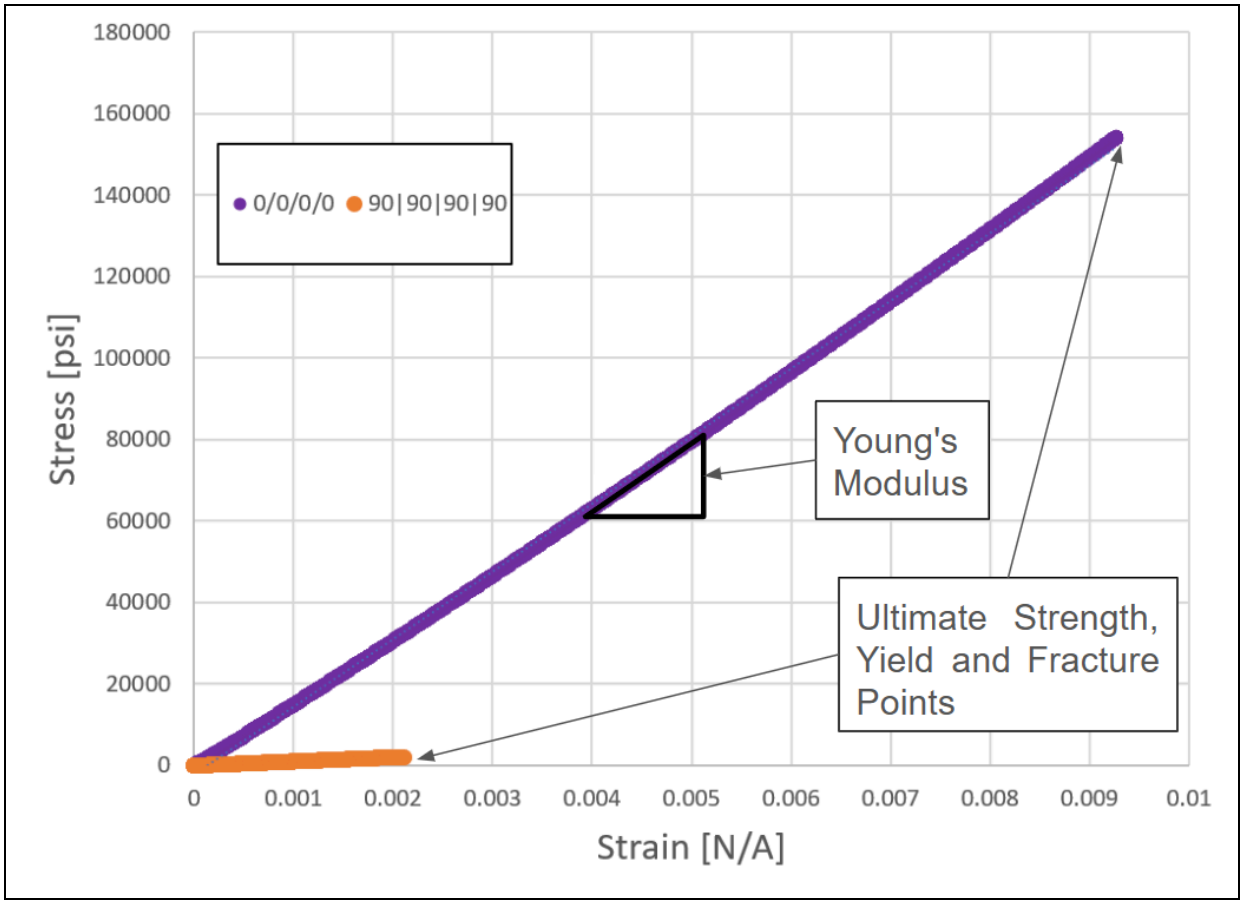


Figure 2: Carbon Fiber Samples Stress vs Strain

4. Analytical Methods

The heart of material analysis is stress and strain. It is through using these that the graphs previously discussed can be created, and thus compared to one another. Stress (1) is calculated using force, given by the testing machine, as well as the cross sectional area, which is standardized by the ASTM.

$$\sigma = \frac{F}{A} \tag{1}$$

Now that the y-axis has been found, strain will be calculated next for the x-axis. Strain is solved by using the change in length over the original length. This gives a unitless number. The change in length is measured by the extensometer which is attached to the sample, and measures the displacement. With displacement values, and the original length, strain can be calculated using equation (2).

$$\varepsilon = \frac{\Delta L}{L} \quad (2)$$

The elastic (linear) region of the graph has a constant relationship between stress and strain. This differs per material, and is called the modulus of elasticity. The relationship can be seen below where σ is stress (1), and ε is strain (2).

$$\sigma = E\varepsilon \quad (3)$$

A statistical analysis is applied in order to create a 99% confidence level in the lower bound. This means that 99% of the values found will fall above this bound. It is calculated using the mean yield strength (σ_y), as well as the (t) value. (t) value is equal to the number of samples minus one, and $s_{\sigma y}$ is the standard deviation.

$$\sigma_{99\%} = \overline{\sigma_y} \pm t_{3,99\%} s_{\sigma_y} \quad (4)$$

Toughness is the ability of a material to resist fracture when under stress. It can be found by taking the area under both the elastic, and plastic regions. This is done using trapezoidal riemann sums, using formula (5).

$$\sum \frac{1}{2} \cdot (\sigma_1 + \sigma_2) \cdot (\varepsilon_2 - \varepsilon_1) \quad (5)$$

The strength of carbon fiber varies with its orientation, which makes it an anisotropic material. Using the rule of mixtures, the modulus of elasticity can be approximated. When all fibers are in 0|0|0|0, the modulus of elasticity can be approximated by equation (6).

$$E_c = E_f V_f + E_m V_m \quad (6)$$

Where E_c is the modulus of elasticity of the composite, E_f is the modulus of elasticity of the fiber, and E_m is the modulus of elasticity of the matrix. V_f and V_m are the respective volumes of the fiber and matrix. If the fibers are oriented in 90|90|90|90, equation (7) can be used.

$$\frac{1}{E_f} = \frac{V_f}{E_f} + \frac{V_m}{E_m} \quad (7)$$

Where all of the variables listed for equation (6) are true for equation (7). In order to calculate the yield point for aluminum, a 0.2% offset line was necessary. This is because there is not one specific yield point, but more of a transformation between elastic, and plastic. The offset line is useful in finding the middle of this region. This was plotted by using the modulus of elasticity (3), as well as the values for strain found earlier, and subtracting 2%. An intercept, (b) was calculated using the original experimental values. This formula can be seen below in equation (8).

$$\varepsilon_{0.2\%} = E(\varepsilon_n - 0.002) + b \tag{8}$$

Specific strength was needed, and solved for by dividing the ultimate strength by the density of the material. This allows for even comparisons to other materials. The formula can be seen below in equation (9).

$$S = \frac{\sigma_{max}}{\rho} \tag{9}$$

Finally, the percent error ($\%_E$) was calculated to estimate the accuracy of the data. This is done by subtracting the experimental values from the calculated values, dividing by calculated values, and multiplying by 100 to make it a percent.

$$\%_e = \left(\frac{Experimental - Calculated}{Calculated}\right) \cdot 100 \tag{10}$$

5. Analytical Results

Using the stress strain plots from Figure 1 and Figure 2, The important values were pulled from the chart, and the modulus of elasticity was calculated using equation (3) along with linear regression in order to ensure the most correct value is found. The location of these values were labeled on Figures 1, and 2.

Table 1

Values from Figure 1

	Young's Modulus [psi]	Ultimate Strength [psi]	Fracture Point [psi]	Toughness [lbf*in/ in^3]	Yield Strength [psi]	Specific Strength [lbf*in/lbm]
Al 6061-T6	8794225.407	44591.216	31751.241	5090.099	40806	447771.9
0|0|0|0	16600674.8	154167.3	154167.3	714.69	154167.3	2904985
90|90|90|90	1002587.1	2130.8	2130.8	2.24	2130.8	59155.81

By analyzing all four samples of aluminum and plotting both their stress strain curve, as well as the 0.2% offset lines, the mean yield point was found. The lower bound was calculated using formula (4).

Table 2

Data from all Aluminum Samples

Mean Yield Point [psi]	Standard Deviation Yield Point [psi]	Lower bound 99% [psi]
41919.99	40447.16	29941.38

Finally, using the rule of mixtures, the theoretical modulus of elasticity could be solved for. By comparing the experimental and theoretical values, a percent error could be calculated.

Table 3

Carbon fiber Rule of Mixtures

Sample	Experimental E [psi]	Theoretical E [psi]	Percent Error [%]
0|0|0|0	16600674.8	20272000	18.11
90|90|90|90	1002587.1	1413189.38	29.06

6. Discussion

Aluminum and carbon fiber are vastly different materials, and the difference of their properties makes them both perfect candidates for comparison. Beginning with table 1, an important basis of

comparison is the modulus of elasticity. This is relevant to the discussion because it represents the stress that a material can withstand per unit of strain. The higher this number, the less a material will stretch when under load. 0|0|0|0 carbon fiber has the highest value, with 90|90|90|90 having the lowest. The 90|90|90|90 carbon fiber was improperly applied, due to carbon fiber gaining most of its strength by being parallel with the force. The 90|90|90|90 carbon fiber has twice the modulus of elasticity as the aluminum sample, which makes it more applicable when tight tolerances are necessary, due to the aluminum expanding more when under the same stress.

Comparing the yield strength of all three samples is important to determine the application. Once a material passes through the yield point, it will begin to deform, and will not be able to return to its pre-stressed state. Similar to the modulus of elasticity, 0|0|0|0 carbon fiber has the highest, with 90|90|90|90 in last. What is notable for the samples is that the yield point and failure point are identical. This gives aluminum the advantage, which can be seen during the discussion of toughness.

Ultimate strength is defined as the maximum stress a material will undergo throughout the testing. For aluminum, it is at the peak of the plastic region (Figure 1), and for the carbon fiber samples, it is the yield point (Figure 2). The ultimate strength is an important factor, because deformation is necessary for the material to perform its job, knowing how much stress a material can withstand is crucial when making design decisions. An example of this could be found in the crash bar in the front bumper of modern cars. Ensuring that the ultimate strength is above the highest possible value will always ensure the part will perform its duty correctly.

Toughness is the area underneath the stress strain curve, and is the ability of a material to resist fracture under stress, and therefore is relatively higher on materials with a plastic region. This is because a material can exit the elastic region, and enter the plastic region without fracturing, like aluminum, trading the stress for permanent deformation instead of fracture. For carbon fiber, which does not have a plastic region, it will suffer a lower toughness. This is shown in the data in Table 1, with aluminum having a much higher value of toughness than either carbon fiber sample.

Finally, the last point of comparison between all three samples is the strength to weight ratio. This is defined as the amount of force a material is capable of resisting per unit of mass. This is important when making decisions about what material to use. Weight restrictions are common, and typically seen in spacecraft and airlines, but can be found anywhere. Due to carbons low atom weight, the 0|0|0|0 possesses a much higher strength to weight ratio than aluminum (Table 1). The 90|90|90|90 sample had a lower ratio than the aluminum, but that will be discussed in a future section. The impressive strength to weight ratio of aluminum is why most airplanes were built from aluminum, and when carbon fiber was made cheaper and more accessible, airplanes have begun to switch away from aluminum and to a material that possesses more strength per unit weight.

Two types of carbon fiber were tested, and both are displayed in Figure 2. Using the method of mixtures, both of their modulus of elasticity were calculated, and compared to their theoretical modulus of elasticity. This, along with the percent error can be seen in Table 3. The percent error can come from less than perfect assembly of the carbon fiber, or uneven heating in the autoclave. As far as the comparisons between the modulus of elasticity, neither of these are the most common type of carbon fiber, which is a 45|45|45|45 weave. The 90|90|90|90 posses relatively zero strength, where the 0|0|0|0 possesses immense strength. These are both the same sample, but under tension from two different angles relative to the direction of the fibers. The 45|45|45|45 weave ensures a more even distribution of force, and the layers weaving between each other further enhance the strength. The direction and alignment of the fibers plays the largest role in the strength of carbon fiber, and knowing where the material will be used, will determine the arrangement of fiber used. If the carbon fiber will only be in tension from opposing directions, 0|0|0|0 is the best option, however if force is expected from a third direction, or the forces are not opposite, 45|45|45|45 is the best way forward.

7. Conclusion

Aluminum and carbon fiber both redefined the world when they were made widely available. They both hold special and unique properties that make them key to understanding the fundamentals of material properties. Material properties is a crucial skill for engineers to learn, and master. Material choice

will either be defined by the job, or define the job. To be able to read, comprehend, and compare the stress strain curves, as well as the data found in tables 1 and 2, will allow the best choices to be made regarding material choice. Material properties are possibly the most important part of engineering, and possessing the greatest level of understanding will lead to the best engineers possible.

8. References

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